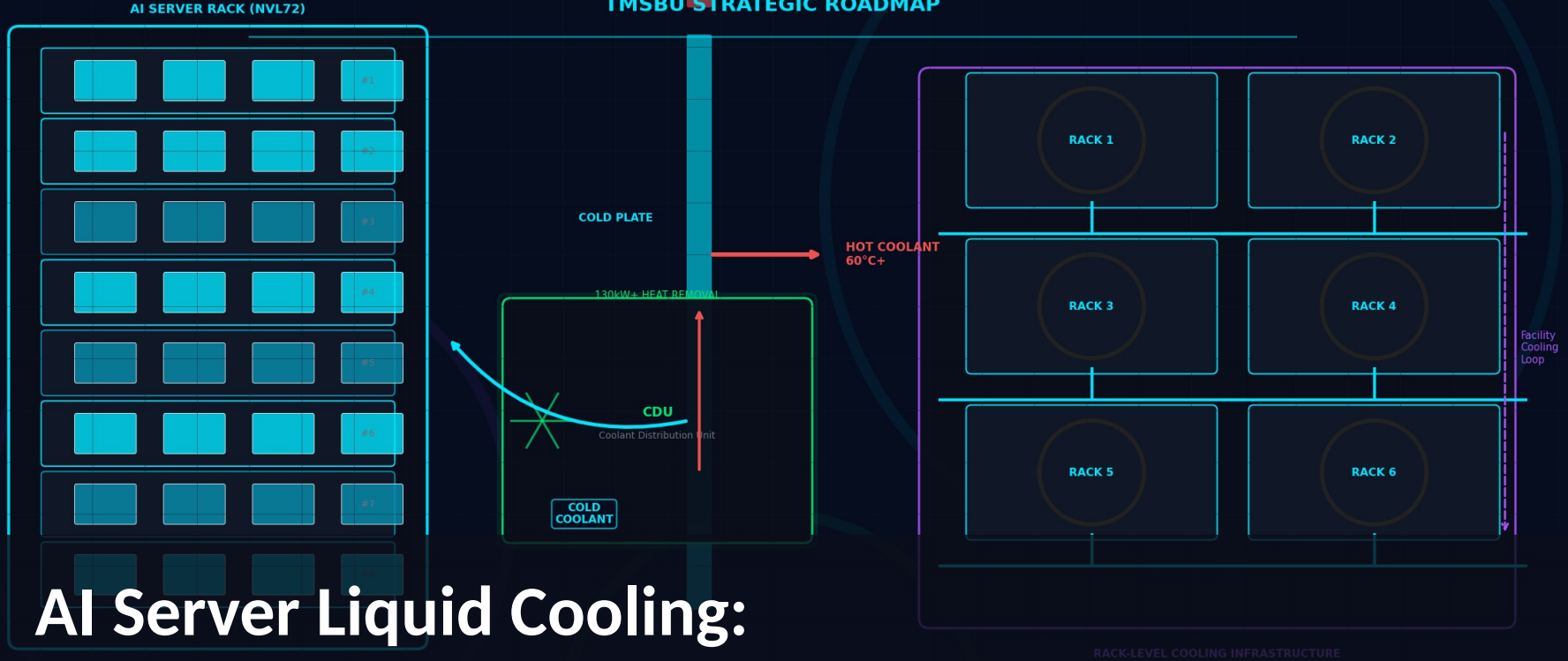


AI SERVER LIQUID COOLING

TMSBU STRATEGIC ROADMAP



AI Server Liquid Cooling:

TMSBU Commercial Opportunity & Strategic Roadmap

V6 — Visual & Content Enhancement

Executive Summary

02

Liquid cooling is not only a product opportunity — it is a business model transition for TMSBU.

01

AI GPU / Rack Power Exceeds Air Cooling

GPU TDP 突破 700W ， GB200 NVL72 rack power 達 130kW+ 。風冷已達物理瓶頸。

02

Liquid Cooling Adoption Accelerating

液冷滲透率：2023 年 20% → 2026 年 60% → 2030 年 80%+ 。

03

Hybrid Air + Liquid Remains Necessary

GPU 本體用液冷， memory 、 PCB 、 power 仍需風冷。兩者長期共存。

04

TMSBU Can Expand into Module & System

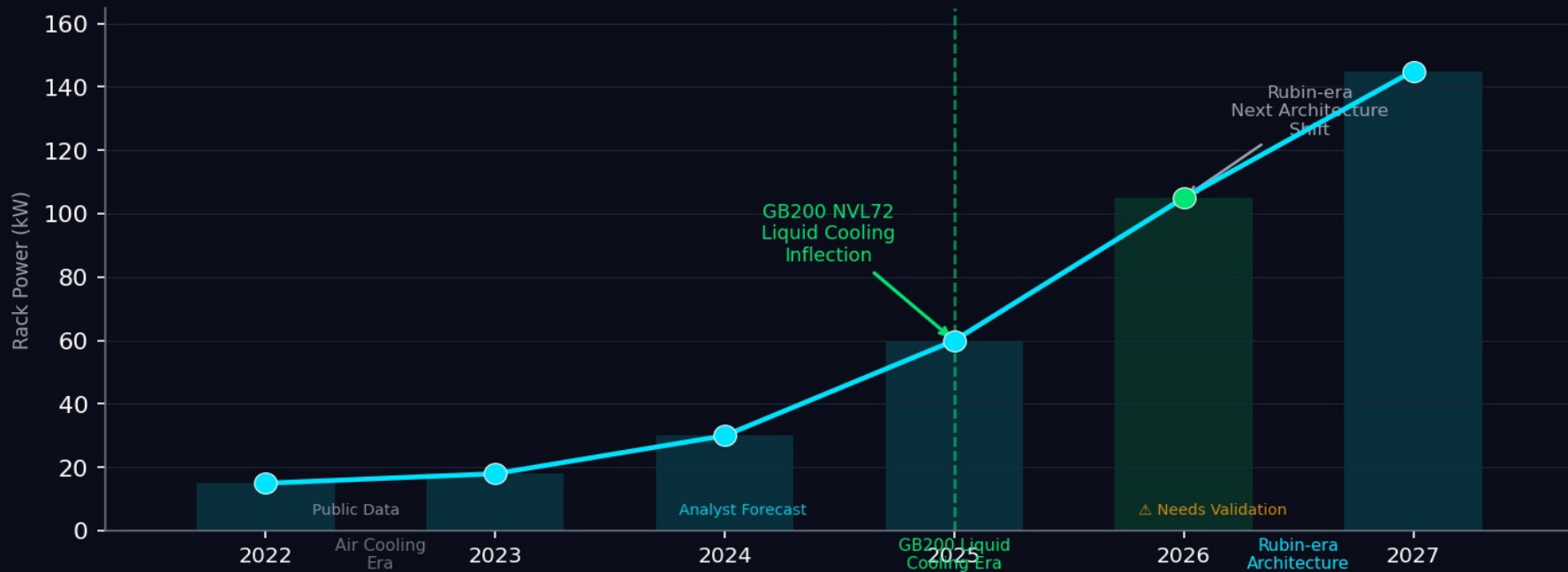
Fan 、 HS 、 HP 、 VC 、 Cold Plate 已有基礎，升級為系統供應商。

05 18 Months Are Critical

Market Inflection: AI Compute Density Hits Thermal Wall

03

液冷在 2025 年 GB200 NVL72 達到爆發點，2027+ Rubin 架構將功率密度推向更高

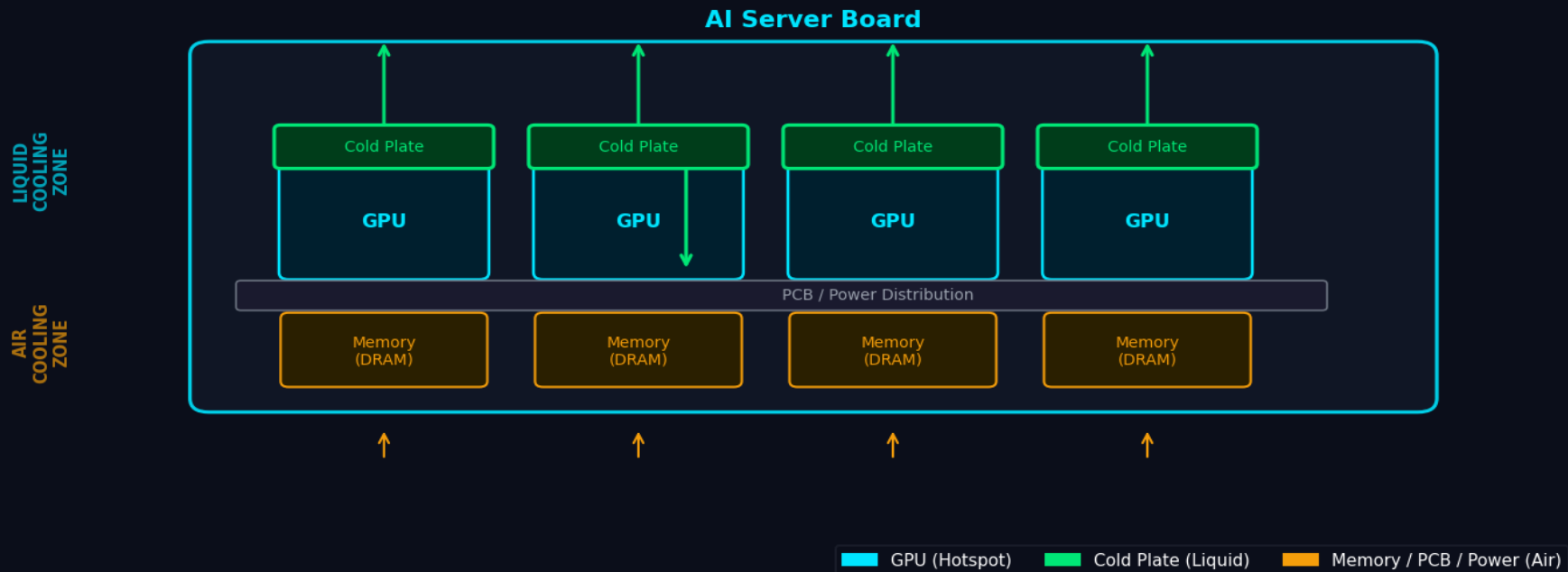


KEY INSIGHTS

- GPU TDP 1,000W+ (next-gen projection) 已是液冷的基本門檻
- Rack power 80kW+ 將在 2026-2027 年成為 hyperscale 主流
- TMSBU 現有 air-cooling 營收在 2025 後將面臨結構性壓力

Hybrid Air + Liquid: Hotspots vs. System-Wide Cooling

液冷處理 GPU 熱點，風冷處理其餘系統組件——兩者長期共存

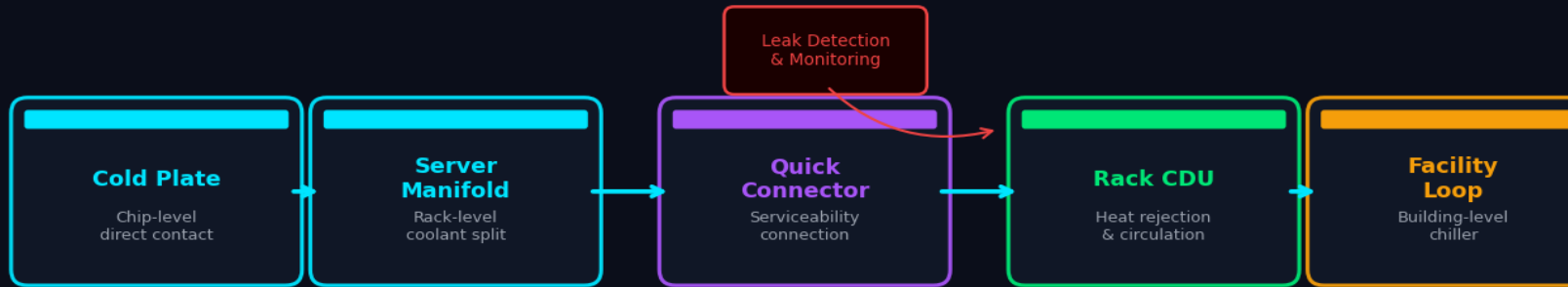


KEY INSIGHTS

- Cold Plate 直接接觸 GPU chip，散熱效率遠優於風扇
- Memory / PCB / Power 仍以氣流散熱為主，不可完全替換
- Hybrid architecture 是 2025-2028 年性價比最優解

Chip-to-Rack Liquid Cooling Architecture

System value comes from integration: cold plates, manifolds, CDUs, connectors, monitoring



Source: Industry projection · Some specs need TMSBU internal validation

Coolant Flow →

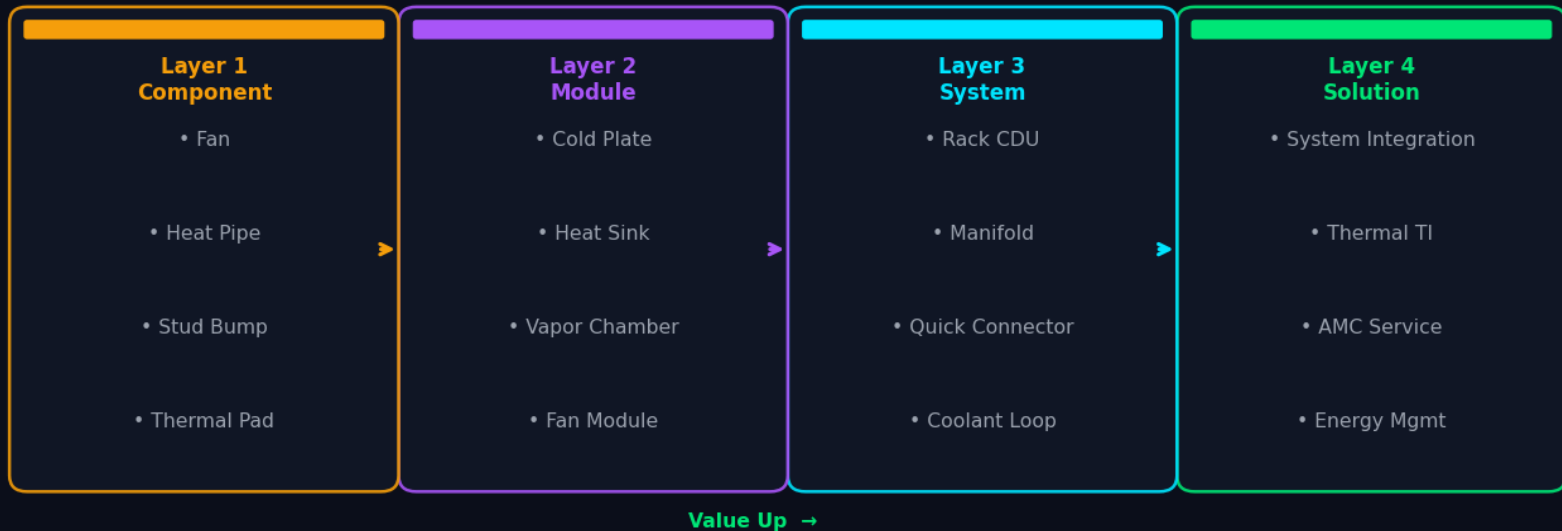
KEY INSIGHTS

- Quick Connector 位於 serviceability 介面，不影響主熱流路徑
- Monitoring / Leak Detection 為附加安全節點，非熱流節點
- CDU capacity Δ 需要 TMSBU 內部驗證 (未使用 130kW+ 等未經數字)

Architecture based on industry-standard direct-to-chip cooling loop. CDU specs Δ require TMSBU internal validation.

TMSBU Portfolio: 4-Layer Value Upgrade Path

Component → Module → System → Solution : 每層疊加更高客戶價值



Delta Group Synergy: Four Quadrant Advantage

07

Power Electronics × Thermal Management × Control Systems × Infrastructure

Power Electronics

- Power supply
- UPS systems
- Energy conversion

Thermal Management

- Cold plates
- Heat sinks
- CDU / manifolds

Control Systems

- BMS / EMS
- Thermal sensors
- Auto control

Infrastructure

- Rack PDU
- Coolant distribution
- Facility integration

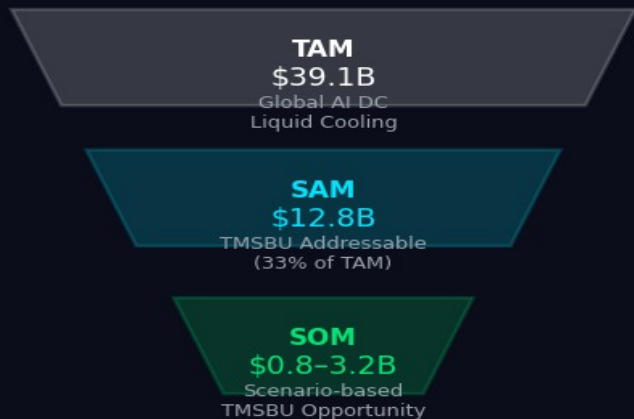
Customer Value: Lower integration risk · Faster design-in · Better reliability · Stronger energy-efficiency support

Market Opportunity: TMSBU's \$0.8B–\$3.2B SOM

08

Scenario-based TAM/SAM/SOM framework — *⚠ SOM requires internal TMSBU pipeline validation*

Market Funnel



⚠ SOM is scenario-based and requires TMSBU internal validation

TMSBU Opportunity Scenarios



⚠ SOM is scenario-based and requires TMSBU internal validation

Capability Gap & Risk Assessment

09

Impact × Urgency matrix — immediate priorities need executive decision

Impact × Urgency Risk Matrix

	Low Urgency	Med Urgency	High Urgency
High Impact		IP / Patent Coverage	Platform Design-in Timing
Med Impact	Component Supply Chain	Thermal Engineer Talent Gap	Chinese Cost Pressure
Low Impact		Customer Qualification	Leak Reliability Qualification

High Risk
Med Risk
Low Risk

TOP 3 IMMEDIATE PRIORITIES

1 CDU Strategy

Build · Partner · Acquire decision required in 90 days

2 Leak Reliability

Qualification process for cold plate + manifold + connector

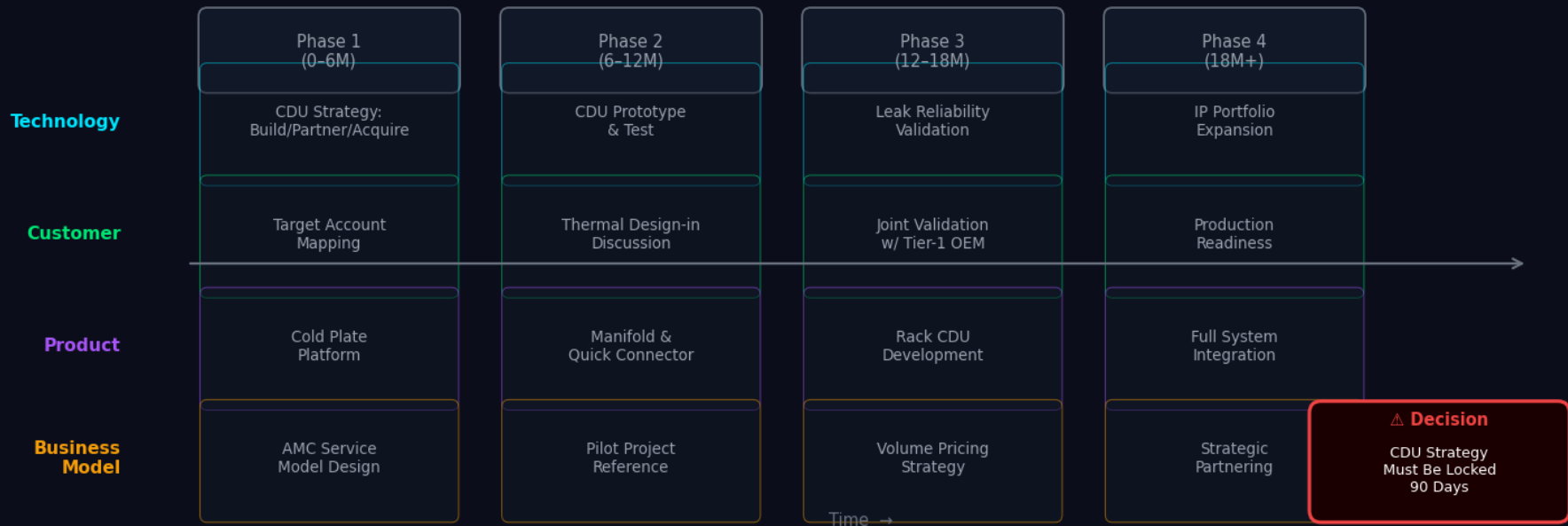
3 Customer Qualification

Tier-1 server OEM/ODM thermal design-in pipeline

Strategic Roadmap: 4-Lane Execution Plan

10

Technology · Customer · Product · Business Model — 6-18 month window



KEY INSIGHTS

- Phase 1 (0-6M) : 確認 CDU 策略，鎖定目標帳戶，展開 Thermal Design-in 討論
- Phase 2 (6-12M) : CDU 原型驗證，與 Tier-1 客戶建立 Joint Validation
- Phase 3 (12-18M) : CDU / Manifold / Quick Connector 產品化，通過可靠性認證